

Binomial Theorem.

Why Binomial Theorem?

In Lower classes, we learned

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

But for higher Powers like $(a+b)^5$ or $(a-b)^6$, multiplying becomes hard.

What Does Binomial Theorem Do?

It gives a formula to expand $(a+b)^n$ easily, where n is a Positive integer.

no. of terms $\rightarrow$ Binomial theorem.
 $(a+b)^8$
 $8+1=9$ terms

$$[a^2 + 2ab + b^2]^4$$

$$[(a^2 + 2ab + b^2)^2]^2$$

$$[(a^2 + 2ab)^2 + 2(a^2)(2ab) + (b^2)^2]^2$$

$$[(a^4)^2 + 2(2ab) + (2ab)^2 + 4a^3b + b^4]^2$$

Important Observation: $(a+b)^2$ has 3 terms
 $(3+1)=4$

1. Total terms in $(a+b)^n = n+1$

2. Powers of a decrease and b increase in each term.

• 1st term: $a^n b^0$
 • 2nd term: $a^{n-1} b^1$
 • ...
 • Last term: $a^0 b^n$

3. In every term: Power of a + Power of b = n

4. Co-efficients follow Pascal's Triangle

$(a+b)^2 =$
 $a^2 b^0$ | $a^{2-1} b^1$ | $a^{2-2} b^2$
 $a^2 b^0$ | $a^1 b^1$ | $a^0 b^2$
 $3C0 a^2$ | $2C1 ab$ | $1C2 b^2$

General form of Binomial Theorem:

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Where:

$\binom{n}{k}$ = nCk (combination)

a and b = terms of the binomial

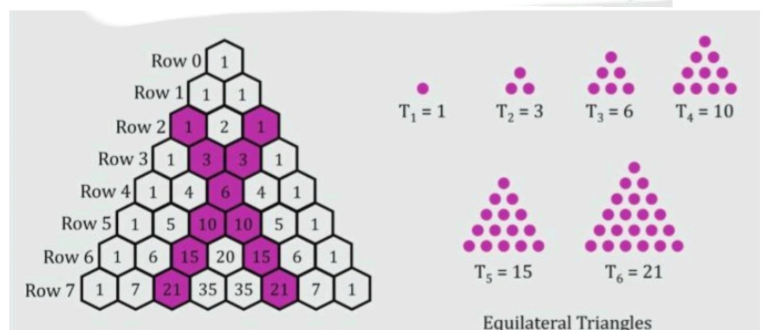
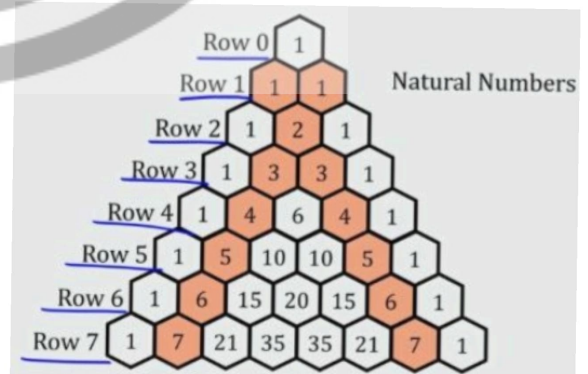
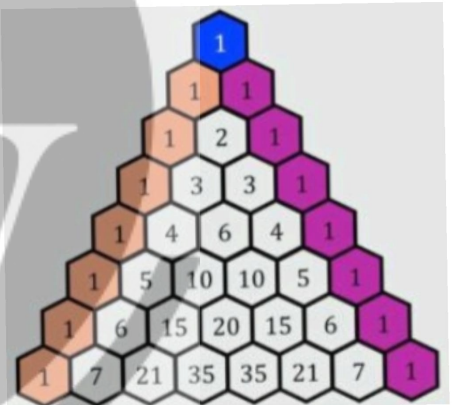
n = Power

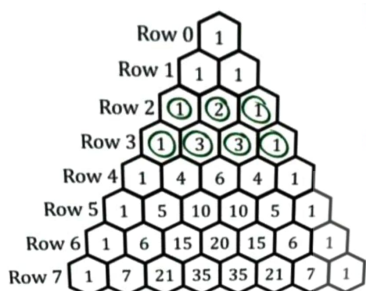
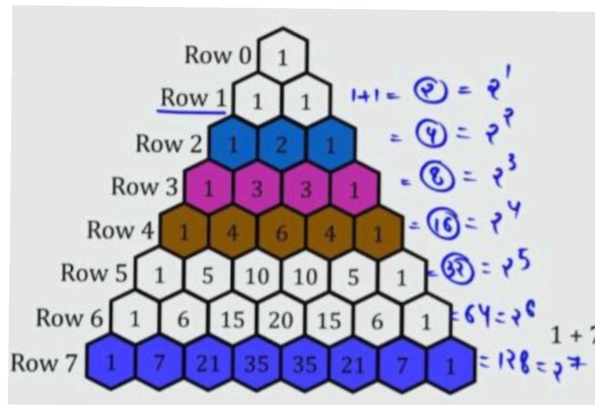
Example:

$$(a+b)^2 = \binom{2}{0} a^2 b^0 + \binom{2}{1} a^1 b^1 + \binom{2}{2} a^0 b^2 = a^2 + 2ab + b^2$$

Pascal's Triangle:

Why we study?
 To understand Binomial Theorem properly.





Row 2: $(a+b)^2 = a^2 + 2ab + b^2$
 Row 3: $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
 Row 4: $(a+b)^4 = 1 \quad 4 \quad 6 \quad 4 \quad 1$

Binomial Theorem for any Positive integer n .

$$(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_{n-1} a^1 b^{n-1} + {}^nC_n a^0 b^n$$

$$(a-b)^n = {}^nC_0 a^n b^0 - {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 - \dots - {}^nC_{n-1} a^1 b^{n-1} + {}^nC_n a^0 b^n$$

* $(a+b)^2 = 2C_0 a^2 b^0 + 2C_1 a^1 b^1 + 2C_2 a^0 b^2$
 $\rightarrow 1 \times a^2 \times 1 + \frac{2}{1} a^1 b^1 + 1 \times 1 \times b^2$

$\rightarrow 1 \times a^2 \times 1 + \frac{2}{1} a^1 b^1 + 1 \times 1 \times b^2$

$\{a^2 + 2ab + b^2\}$ Ans.

TRICK

$3C_2 \Rightarrow \frac{3 \times 2}{2 \times 1}$	$8C_2 \Rightarrow \frac{8 \times 7}{2 \times 1}$
$6C_3 \Rightarrow \frac{6 \times 5 \times 4}{3 \times 2 \times 1}$	$10C_1 \Rightarrow \frac{10}{1}$

$\{nC_0 = nC_n = 1\}$

Particular Cases of Binomial Theorem —

① $(a-b)^n$

Just like $(a+b)^n$, but signs change (Plus, minus, Plus, minus)

The sign of each term depends on Power of b :

Sign = $(-1)^x$

So:

- Where x is even $\rightarrow$ term is Positive
- Where x is odd $\rightarrow$ term is Negative.

Example:

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

(Notice the signs: + - + -).

② $(a+b)^n + (a-b)^n$

Formula:

- $(a+b)^n + (a-b)^n = 2 \times (\text{Even-Powered terms only})$.
- All odd-Powered terms cancel.

Example:

Let's take $(a+b)^2 + (a-b)^2$
 $= (a^2 + 2ab + b^2) + (a^2 - 2ab + b^2) = 2a^2 + 2b^2 = 2(a^2 + b^2)$

Only even Powers (like a^2, b^2) remain.

③ $(a+b)^n - (a-b)^n$

Formula:

- $(a+b)^n - (a-b)^n = 2 \times (\text{Odd-Powered terms only})$
- All even-Powered terms cancel.

Example:

Take $(a+b)^3 - (a-b)^3$
 $= (a^3 + 3a^2b + 3ab^2 + b^3) - (a^3 - 3a^2b + 3ab^2 - b^3)$
 $= 6a^2b + 6ab^2 = 6ab(a+b)$

Only odd-Powered terms remain and are doubled.

④ $(1+x)^n$

• Just put $a=1, b=x$

• $(1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$

Example:

$(1+x)^3 = 1 + 3x + 3x^2 + x^3$

⑤ $(1-x)^n$

• Put $a=1, b=-x$

• $(1-x)^n = \binom{n}{0}x^0 - \binom{n}{1}x^1 + \binom{n}{2}x^2 - \dots + (-1)^n \binom{n}{n}x^n$

Example:

$(1-x)^3 = 1 - 3x + 3x^2 - x^3$

1. $(a+b)^n = nC_0 a^n b^0 + nC_1 a^{n-1} b^1 + nC_2 a^{n-2} b^2 + \dots + nC_n a^0 b^n$

2. $(a-b)^n = nC_0 a^n b^0 - nC_1 a^{n-1} b^1 + nC_2 a^{n-2} b^2 - \dots + nC_n a^0 b^n$

$(a-b)^4 = 4C_0 a^4 b^0 - 4C_1 a^3 b^1 + 4C_2 a^2 b^2 - 4C_3 a b^3 + 4C_4 a^0 b^4$
 $= 1 \cdot a^4 \cdot 1 - \frac{4 \times 3}{2 \times 1} a^3 b + \frac{4 \times 3 \times 2}{3 \times 2 \times 1} a^2 b^2 - 1 \times 1 \times b^4$

$(a-b)^4 = a^4 - 4a^3 b + 6a^2 b^2 - 4ab^3 + b^4$

* $8C_3$
 $\hookrightarrow \frac{8 \times 7 \times 6}{3 \times 2 \times 1}$

* $10C_6$
 $\hookrightarrow \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$

$\Rightarrow 30 \times 7$

$\Rightarrow 210$

Example) $nC_0 = 1 \Rightarrow 5C_0 = 1$
 $nC_n = 1 \Rightarrow 5C_5 = 1$

Expand each of the Expression.

$(1-2x)^5$ no. of Terms $\Rightarrow (n+1)$
 $\Rightarrow 5+1$
 $= 6$

$\hookrightarrow 5C_0 1^5 (2x)^0 - 5C_1 1^4 (2x)^1 + 5C_2 1^3 (2x)^2 - 5C_3 1^2 (2x)^3 + 5C_4 1^1 (2x)^4 - 5C_5 1^0 (2x)^5$
 $\hookrightarrow 1 \cdot 1^5 \cdot 1 - 5 \cdot 1^4 \cdot 2x + \frac{5 \times 4}{2 \times 1} \cdot 1^3 \cdot 4x^2 - \frac{5 \times 4 \times 3}{3 \times 2 \times 1} \cdot 1^2 \cdot 8x^3 + \frac{5 \times 4 \times 3 \times 2}{4 \times 3 \times 2 \times 1} \cdot 1^1 \cdot 16x^4 - 1 \cdot 1 \cdot 32x^5$

$\Rightarrow 1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5$

Pth Term from the End:

[Pth term for the end = $(n-P+2)^{th}$ term from the beginning.]

Use this when asked for a term from the end of the expansion.

Example:

Find the 4th term from the end in $(x+2)^7$.

• $n=7, P=4$

• So, 4th from end = $(7-4+2)^{th}$ = 5th term

• $T_5 = \binom{7}{4} x^3 (2)^4 = 35 \cdot x^3 \cdot 16 = 560x^3$

4th term from the $(x+2)^7$ end

$\oplus (a+b)^3$
 $\hookrightarrow a^3 + 3a^2b + 3ab^2 + b^3$

Term = $nC_r a^{n-r} b^r$

$(T_4) \rightarrow T_{3+1}$
 $T(n+r)$

Pth Term from the End:

Middle Term(s):

Depends on whether n is even or odd.

If n is even:

Total terms: $n+1$ (odd)

Only one middle term = $\left(\frac{n}{2} + 1\right)^{th}$ term

If n is odd:

Total terms = $n+1$ (even)

Two middle terms = $\left(\frac{n+1}{2}\right)^{th}$ & $\left(\frac{n+3}{2}\right)^{th}$ term.

Middle Terms in Binomial Expansion:

Case - (1):

When n is even

Total terms = $n+1$ (which is odd)
So, there will be only one middle term.

Middle Term = $(n/2 + 1)^{\text{th}}$ term

Example:

Find the middle term of $(a+b)^6$

Hence, $n=6$ (even)

Middle Term = $(6/2 + 1) = 4^{\text{th}}$ term

So, 4th term is the middle term.

Case - (2):

When n is odd

Total terms = $n+1$ (which is even)
So, there will be two middle terms

Middle terms =

- $(n+1)/2^{\text{th}}$ term
- $(n+3)/2^{\text{th}}$ term

Example:

Find the middle term of $(a+b)^5$

Here, $n=5$ (odd)

Middle Term:

- $(5+1)/2 = 3^{\text{th}}$ term
- $(5+3)/2 = 4^{\text{th}}$ term

So, 3th & 4th terms are the middle terms.

Special Results:

Sum of All coefficients:

If $a=1, b=1$ then

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$$

Sum of Even & Odd Coefficients:

If $a=1, b=-1$ then

$$\binom{n}{0} + \binom{n}{2} + \dots = \binom{n}{1} + \binom{n}{3} + \dots = 2^{n-1}$$

Both odd and even positioned coefficients are equal.

General Term of an Expansion:

$$(a+b)^n$$

$$T_{r+1} = {}^nC_r (a)^{n-r} (b)^r$$

Q. Find the 4th term from the expansion $(x+2)^7$

$$\therefore r=3, n=7$$

$$T_{3+1} = {}^7C_3 (x)^{7-3} (2)^3 \Rightarrow T_{4+1}$$

$$= \frac{7 \times 6 \times 5}{3 \times 2 \times 1} (x)^4 \times 8$$

$$T_4 = 280x^4$$

Quest) $(x+2)^7$ find 3rd term from the last in the expansion

Beginning $\Rightarrow (n-r+2)$

$$\Rightarrow (7-3+2)$$

$$\begin{matrix} r=5 \\ n=7 \end{matrix}$$

Starting $\Rightarrow (r)$

$$T_6 = T_{5+1} = {}^nC_r (a)^{n-r} (b)^r$$

$$= {}^7C_5 (x)^{7-5} (2)^5$$

$$= \frac{7 \times 6 \times 5 \times 4 \times 3}{5 \times 4 \times 3 \times 2 \times 1} \cdot (x)^2 \cdot 32$$

$$= 21 \times 32 \times x^2$$

$$= 672x^2$$

Quest) Find $(a+b)^4 - (a-b)^4$. Hence evaluate $(\sqrt{3}+\sqrt{2})^4 - (\sqrt{3}-\sqrt{2})^4$.

$$(a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4) - (a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4)$$

$$\Rightarrow a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 - a^4 + 4a^3b - 6a^2b^2 + 4ab^3 - b^4$$

$$\Rightarrow 4a^3b + 4ab^3 + 4a^3b + 4ab^3$$

$$\Rightarrow 8a^3b + 8ab^3$$

$$\Rightarrow 8ab(a^2 + b^2)$$

$$(a+b)^4 - (a-b)^4 = 8ab(a^2 + b^2)$$

$$(\sqrt{3}+\sqrt{2})^4 - (\sqrt{3}-\sqrt{2})^4 = 8(\sqrt{3})(\sqrt{2})[(\sqrt{3})^2 + (\sqrt{2})^2]$$

$$= 8\sqrt{6}[3+2]$$

$$= 8\sqrt{6} \times 5$$

$$= 40\sqrt{6} \text{ Ans (final)}$$

$$\textcircled{+} (a+b)^4$$

$$= 4C_0(a)^4(b)^0 + 4C_1(a)^3(b)^1 + 4C_2(a)^2(b)^2 + 4C_3(a)^1(b)^3 + 4C_4(a)^0(b)^4$$

$$\Rightarrow (1 \cdot a^4 \cdot 1) + 4a^3 \cdot b + \frac{4 \times 3}{2 \times 1} a^2 \cdot b^2 + \frac{4 \times 3 \times 2}{3 \times 2} a \cdot b^3 + 1 \cdot 1 \cdot b^4$$

$$\Rightarrow \{a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4\}$$

$$\textcircled{+} (a-b)^4 = +a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4$$

Example)

Show that $9^{n+1} - 8^n - 9$ is divisible by 64, whenever n is a positive integer.

$$9^{n+1} = (8+1)^{n+1}$$

$$= {}^{n+1}C_0(8)^{n+1}(1)^0 + {}^{n+1}C_18^n(1)^1 + \dots + {}^{n+1}C_{n-1}8^2(1)^{n-1} + {}^{n+1}C_n8 + {}^{n+1}C_{n+1}$$

$$= 64(k) + (n+1)8 + 1$$

$$9^{n+1} = 64k + 8n + 8 + 1$$

$$9^{n+1} - 8n - 9 = 64k$$

Hence, $9^{n+1} - 8n - 9$ is divisible by 64.

$$\begin{aligned} {}^{n+1}C_n &= \frac{(n+1)!}{1!n!} \\ &= \frac{(n+1)n!}{(n-)!} \\ &= n+1. \end{aligned}$$